

Systematic Investment Architecture

From Signals to Strategies to Portfolios

Prof. Mike Aguilar

Economics Department and Fuqua School of Business, Duke University

<https://www.linkedin.com/in/mike-aguilar-econ>

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Outline

1 Overview

2 Signals

- Defining Signals
- Evaluating Signals
 - Single Hold-Out Designs
 - Multiple Hold-Out Designs

3 Strategy

- Timing
- Position Sizing
- Strategy Evaluation

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Motivation

- These notes accompany the Duke Economics course *Machine Learning for Finance*.
- The course focuses on how machine learning methods can be used to design, evaluate, and implement systematic investment strategies.
- The goal of this document is to lay out a clear, modular architecture for a systematic investment process.
- Many practical considerations (e.g., transaction costs, slippage, execution details) are intentionally set aside so we can focus on the components most relevant for illustrating machine learning ideas.
- The framework is designed to be general enough to accommodate a wide range of signals, strategies, and portfolios. Throughout, a simple momentum example is used to make concepts concrete.

Q: Signal vs Strategy vs Portfolio?

A:

- Signal:
- Strategy:
- Portfolio:

Portfolio Creation Process



- t : highest frequency at which portfolio can be traded (e.g. end of week, end of day, intraday);
- $X_{i,t}$: characteristics of asset i at time t , e.g. returns, volume, sector
- Z_t : general features, e.g. characteristics of assets other than asset i , macro state variables
- $s_{i,t}$: signal $\in \{-1, 0, 1\}$ defining short, neutral, or long position
- $s_{i,t}^{pos}$: position $\in \{-1, 0, 1\}$ implied by signal after timing and exit rules
- $w_{i,t}^{raw}$: signed pre-normalized position sizes; e.g. proportional to signal strength
- $w_{i,t}^{scaled}$: position weights scaled to enforce strategy constraints such as fully invested
- $w_{i,t}^P$: optimal weighting across potentially multiple strategies and global constraints

A Toy Momentum Signal

Position sizes scaled by momentum magnitude

Stage	Definition / Rule	Comment
t	Daily (end of day)	Trading frequency
$X_{i,t}$	$mom_{i,t}$	Past-day return (momentum)
Z_t	—	No contextual variables
$s_{i,t}$	$\begin{cases} +1 & \text{if } mom_{i,t} > 0 \\ 0 & \text{if } mom_{i,t} = 0 \\ -1 & \text{if } mom_{i,t} < 0 \end{cases}$	Raw directional signal
$s_{i,t}^{\text{pos}}$	$= s_{i,t}$	Position (no timing adjustment)
$w_{i,t}^{\text{raw}}$	$= s_{i,t}^{\text{pos}} mom_{i,t} $	Size by momentum magnitude
$w_{i,t}^{\text{scaled}}$	$\frac{w_{i,t}^{\text{raw}}}{\sum_{j=1}^N w_{j,t}^{\text{raw}} }$	Normalize total abs. exposure
$w_{i,t}^p$	$= w_{i,t}^{\text{scaled}}$	Final portfolio weights

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Defining Signals

Using logical conditions for long, short, neutral positions

Define a directional signal using opposing predicates:

$$s_{i,t} = \mathbf{1}[p^{(+)}(X_{i,t}, Z_t)] - \mathbf{1}[p^{(-)}(X_{i,t}, Z_t)], \quad s_{i,t} \in \{-1, 0, +1\}$$

- $p^{(+)}$: condition for a long signal
- $p^{(-)}$: condition for a short signal
- $s_{i,t} = +1$: long condition true, short false
- $s_{i,t} = -1$: short condition true, long false
- $s_{i,t} = 0$: neither (no signal) or both (conflict) true $\Rightarrow$ neutral

Signal A

Define mom_i as momentum on asset i [ignore the precise definition of momentum for now]

$$s_{i,t} = \mathbf{1}[p^{(+)}(X_{i,t}, Z_t)] - \mathbf{1}[p^{(-)}(X_{i,t}, Z_t)]$$

$$p^{(+)} = mom_{i,t-1} > 0$$

$$p^{(-)} = mom_{i,t-1} < 0$$

Q: Can you explain this signal?

A:

-
-
-
-

Signal B

Q: Can you describe this signal?

$$s_{i,t} = \mathbf{1}[p^{(+)}(X_{i,t}, Z_t)] - \mathbf{1}[p^{(-)}(X_{i,t}, Z_t)]$$
$$p^{(+)} = (\text{industry}_i = \text{Technology})$$
$$p^{(-)} = \text{False}$$

A:



Signal C

Q: Can you describe this signal?

$$s_{i,t} = \mathbf{1}[p^{(+)}(X_{i,t}, Z_t)] - \mathbf{1}[p^{(-)}(X_{i,t}, Z_t)]$$

$$p^{(+)} = (\text{industry}_i = \text{Technology}) \wedge (\text{PE}_{i,t} < \overline{\text{PE}}_{i,\text{hist}})$$

$$p^{(-)} = \text{False}$$

A:



Signal D

Q: Can you describe this signal?

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$
$$p^{(+)} = [\text{PE}_{i,t} < \overline{\text{PE}}_t]$$
$$p^{(-)} = [\text{PE}_{i,t} > \overline{\text{PE}}_t]$$

A:



Signal E

Q: Can you describe this signal?

Forecast returns via $\hat{r}_{i,t+1} = AR(2)(X_{i,t})$

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right],$$

$$p^{(+)}(X_{i,t}, Z_t) \equiv (\text{rank}_t(\hat{r}_{i,t+1}) \leq 10),$$

$$p^{(-)}(X_{i,t}, Z_t) \equiv (\text{rank}_t(\hat{r}_{i,t+1}) > N_t - 10).$$

Forecasts are ranked from highest to lowest, so rank 1 denotes the highest forecasted return.

A:

-
-
-
-

Signal F

Q: Can you describe this signal?

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right],$$

$$p^{(+)}(X_{i,t}, Z_t) \equiv (\text{industry}_i \in \{\text{Tech}, \text{Comm}\}) \wedge (\text{mom}_{i,t-1} > \overline{\text{mom}}_t) \wedge (\text{VIX}_t \geq 25)$$

$$p^{(-)}(X_{i,t}, Z_t) \equiv (\text{industry}_i \in \{\text{Tech}, \text{Comm}\}) \wedge (\text{mom}_{i,t-1} > \overline{\text{mom}}_t) \wedge (\text{VIX}_t \leq 20)$$

A:

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-
-
-

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Sampling Danger

Q: What's the danger in evaluating this toy momentum signal?

- Construct Signal: Compute momentum on stock A for Dec2025 as cumulative return from Jan2025 through Dec2025. Repeat for stocks B,C,D,E.
- Evaluate Signal: Compare Dec2025 returns for those with positive momentum (A,B) to those with negative momentum (C,D,E).

A:

- Time series:
- Cross Section:

Test (aka Hold-Out) vs Train

Using Test/Train Samples to reduce the dangers in the toy momentum signal

- Time Series:
 - Train: Construct momentum on Jan through Nov
 - Test:
- Cross Section:
 - Train: Construct momentum on stocks A,B,C,D,E
 - Test:
- Panel:
 - Train: Construct momentum on stocks A,B,C,D,E using Jan-Nov
 - Test:

Note: We can refer to these three as Single Hold-Out Designs. In each, we are evaluating on a single period and/or single set of assets. In later sections we'll explore Multiple Hold-Out Designs.

Note: The classic 12-2 momentum signal ala Jegadeesh and Titman (1993) goes a step further and defines Dec25 (t) momentum as returns Dec24 ($t-12$) through Oct25 ($t-2$), then tests on Jan26 and beyond

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Evaluating the Signal with Future Returns

Toy Momentum Example

- Define asset-specific momentum during the training window as

$$\text{mom}_i^{\text{Train}}$$

- Construct the signal

$$s_i^{\text{Train}} = \begin{cases} +1 & \text{if } \text{mom}_i^{\text{Train}} > 0, \\ -1 & \text{if } \text{mom}_i^{\text{Train}} < 0, \\ 0 & \text{if } \text{mom}_i^{\text{Train}} = 0 \end{cases}$$

- Evaluate predictive ability: compare test-period returns

$$r_i^{\text{Test}} \mid s_i^{\text{Train}} = +1 > r_i^{\text{Test}} \mid s_i^{\text{Train}} = -1$$

Difference in Means

$$H_0 : \mathbb{E}[r_i^{\text{Test}} \mid s_i^{\text{Train}} = +1] \leq \mathbb{E}[r_i^{\text{Test}} \mid s_i^{\text{Train}} = -1]$$

$$H_1 : \mathbb{E}[r_i^{\text{Test}} \mid s_i^{\text{Train}} = +1] > \mathbb{E}[r_i^{\text{Test}} \mid s_i^{\text{Train}} = -1]$$

$$T = \frac{\frac{1}{N_{\text{Long}}} \sum_{i: s_i^{\text{Train}} = +1} r_i^{\text{Test}} - \frac{1}{N_{\text{Short}}} \sum_{i: s_i^{\text{Train}} = -1} r_i^{\text{Test}}}{\sqrt{\frac{\hat{\sigma}_{\text{Long}}^2}{N_{\text{Long}}} + \frac{\hat{\sigma}_{\text{Short}}^2}{N_{\text{Short}}}}}$$

$$\hat{\sigma}_{\text{Long}}^2 = \frac{1}{N_{\text{Long}} - 1} \sum_{i: s_i^{\text{Train}} = +1} \left(r_i^{\text{Test}} - \frac{1}{N_{\text{Long}}} \sum_{j: s_j^{\text{Train}} = +1} r_j^{\text{Test}} \right)^2$$

- Define σ_{Short}^2 analogously
- If reject the Null, then

Distributional Comparisons

Compare Test-period return distributions conditional on the Train-period signal

- **Signal frequencies and coverage:**

$$N_+, N_0, N_-, \quad \Pr(s^{Train} = +1), \Pr(s^{Train} = 0), \Pr(s^{Train} = -1)$$

$$\text{Coverage} = \Pr(s^{Train} \neq 0) = \frac{N_+ + N_-}{N}$$

- **Conditional return summaries:** compare empirical moments and quantiles across

$$r^{Test} \mid s^{Train} = +1, \quad r^{Test} \mid s^{Train} = 0, \quad r^{Test} \mid s^{Train} = -1$$

for each signal category with sufficient observations.

- **Visual diagnostics:**

- Compare boxplots or histograms across signal groups.
- Overlay kernel densities to examine differences in location, dispersion, skewness, and tails.

- **Nonparametric distributional test:**

- Kolmogorov–Smirnov: test whether two conditional return distributions differ anywhere in their empirical CDFs.

Information Coefficient

Cross-sectional linear association between signal and forward returns

$$IC = \text{Corr}_i(s_i^{Train}, r_i^{Test}) = \frac{\widehat{\text{Cov}}(s^{Train}, r^{Test})}{\hat{\sigma}_s^{Train} \hat{\sigma}_r^{Test}}$$

- s_i^{Train} : signal for asset $i \in \{-1, 0, +1\}$
- r_i^{Test} : forward return in the testing period
- $\widehat{\text{Cov}}(s^{Train}, r^{Test}) = \frac{1}{N} \sum_i (s_i^{Train} - \bar{s}^{Train})(r_i^{Test} - \bar{r}^{Test})$

Information Coefficient (Acted-Only)

Correlation restricted to assets with active signals ($s \in \{-1, +1\}$)

$$IC^{\text{acted}} = \frac{(\bar{r}^{\text{Test}} | s^{\text{Train}} = +1) - (\bar{r}^{\text{Test}} | s^{\text{Train}} = -1)}{\hat{\sigma}_{r^{\text{Test}}} \sqrt{p(1-p)}}$$

- $\bar{r}^{\text{Test}} | s^{\text{Train}} = +1, \bar{r}^{\text{Test}} | s^{\text{Train}} = -1$: mean test-period returns for long and short groups
- $\hat{\sigma}_{r^{\text{Test}}}$: standard deviation of testing period returns on the acted subset
- $p = \frac{N_+}{N_+ + N_-}$: proportion of longs in the acted subset
- Equivalent to the Pearson correlation between $\{s_i^{\text{Train}}, r_i^{\text{Test}}\}$ for $s_i^{\text{Train}} \in \{+1, -1\}$

Testing the Information Coefficient

Assessing significance of IC or IC^{acted}

Parametric correlation test

$$H_0 : IC = 0 \quad H_A :$$

$$t = \frac{IC\sqrt{N-2}}{\sqrt{1-IC^2}}, \quad t \sim t_{N-2}.$$

- N : number of assets (use N_{act} for acted-only case)
- Assumes $r^{Test}|_{S^{Train}}$ approximately Normal
- Equivalent to a difference-in-means test for binary signals

Resampling alternatives (more robust to non-normality and small N)

- **Bootstrap:** resample asset pairs $(s_i^{Train}, r_i^{Test})$, recompute IC, build Confidence Interval.
- **Permutation:** shuffle signal labels across assets to form an empirical null:
 - Recompute IC for each shuffle to build the null distribution.
 - Compare observed IC to this distribution $\Rightarrow$ empirical p -value.

Confusion Matrix

Long vs Not Long (i.e. Short or Neutral)

	$s^{Train} = +1$	$s^{Train} = \{-1, 0\}$
$r^{Test} > 0$	True Positive (TP)	False Negative (FN)
$r^{Test} < 0$	False Positive (FP)	True Negative (TN)

Each cell of the table is frequency of observations meeting that criterion.

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$
- $Precision = \frac{TP}{TP+FP}$
- $Recall = \frac{TP}{TP+FN}$
- $Specificity = \frac{TN}{TN+FP}$
- Type 1 Error = $\frac{FP}{FP+TN}$
- Type 2 Error = $\frac{FN}{TP+FN}$

Confusion Matrix

Q: Can you interpret each of the metrics?

	$s^{Train} = +1$	$s^{Train} = \{-1, 0\}$
$r^{Test} > 0$	True Positive (TP)	False Negative (FN)
$r^{Test} < 0$	False Positive (FP)	True Negative (TN)

Each cell of the table is frequency of observations meeting that criterion.

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$
- $Precision = \frac{TP}{TP+FP}$
- $Recall = \frac{TP}{TP+FN}$
- $Specificity = \frac{TN}{TN+FP}$
- Type 1 Error = $\frac{FP}{FP+TN}$
- Type 2 Error = $\frac{FN}{TP+FN}$

Confusion Matrix

Q: Can you create and interpret a Confusion Matrix for Shorts?

	$s^{Train} = -1$	$s^{Train} = \{+1, 0\}$
$r^{Test} < 0$	True Positive (TP)	False Negative (FN)
$r^{Test} > 0$	False Positive (FP)	True Negative (TN)

Each cell of the table is frequency of observations meeting that criterion.

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$
- $Precision = \frac{TP}{TP+FP}$
- $Recall = \frac{TP}{TP+FN}$
- $Specificity = \frac{TN}{TN+FP}$
- $Type\ 1\ Error = \frac{FP}{FP+TN}$
- $Type\ 2\ Error = \frac{FN}{TP+FN}$

Defining the Hit Rate

Q: What is the meaning of "Hit" Rate?

A:

HitRate =

HitRate^{acted} =

Testing the Hit Rate

Does directional accuracy differ from random guessing?

Large-sample z-test (Binomial Normal Approximation)

$$H_0 : p = 0.5 \quad H_A : p \neq 0.5$$

$$z = \frac{\hat{p} - 0.5}{\sqrt{p_0(1 - p_0)/N}}, \quad p_0 = 0.5, \quad z \sim N(0, 1) \text{ for large } N$$

- $\hat{p}$ = observed hit rate (use $\text{HitRate}_{\text{acted}}$ because the $p_0 = 0.5$ benchmark applies naturally to binary long/short directional predictions).
- Under H_0 , number of hits $K \sim \text{Binomial}(N, 0.5)$.
- Exact binomial test preferred for small N or low coverage.
- In time-series or panel settings, dependence may inflate significance; use HAC or block bootstrap as robustness checks.

Reject H_0 at 5% if Always report coverage and effective sample size.

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Cross Validation (CV)

Robustness comes from many hold-outs, not just one

- **Time Series CV (Walk-Forward)**

- Train: Dec'24 – Nov'25 Test:
- Train: Jan'25 – Dec'25 Test:
- ...

- **Cross-Sectional k -Fold CV (Assets)**

- Example (*illustrative*, $k = 3$):
 - Train: A,B,C,D,E,F Test: G,H
 - Train: A,B,C,D,G,H Test:
 - Train: E,F,G,H Test:
- Test groups are disjoint; each asset appears once as test data
- In practice, choose k so that N/k is large enough to evaluate IC, Hit Rate, and AUC
- Note: We can combine Time Series and Cross Section CV

Creating a Long–Short Signal Portfolio

Q: How form a long short portfolio to illustrate the efficacy of the signal?

Define the long–short portfolio return for test period t as....

...

where N_t^{Long} and N_t^{Short} denote the number of long and short positions, and $s_{i,t}^{Train} \in \{-1, 0, +1\}$ is the signal generated from the training window preceding the test period t .

Assessing the Long–Short Signal Portfolio

- **Seasonality:**
- **Empirical density:**
- **Downside:**

Rolling Information Coefficient

Summarize cross-sectional IC per test window (both full and acted-only)

$$IC_t = \text{Corr}_i(s_{i,t}^{Train}, r_{i,t+h}^{Test}) \text{ and } IC_t^{\text{acted}} = \text{Corr}_{i: s_{i,t}^{Train} \neq 0}(s_{i,t}^{Train}, r_{i,t+h}^{Test})$$

- Newey–West t-stat for mean IC:

$$t_{NW} = \frac{\overline{IC}}{\sqrt{\widehat{Var}_{NW}(\overline{IC})}} \quad (\text{use analogous formula for } \overline{IC}^{\text{acted}}).$$

- Autocorrelation and half-life (use series $\{IC_t\}$ or $\{IC_t^{\text{acted}}\}$):

$$\rho_1 = \text{corr}(IC_t, IC_{t+1}), \quad \text{Half-life} = -\frac{\ln 2}{\ln |\rho_1|}.$$

Summarizing Rolling IC

Statistic	IC (full)	IC (acted-only)
Mean	$\overline{IC}$	$\overline{IC}^{\text{acted}}$
Std. dev.	$\hat{\sigma}_{IC}$	$\hat{\sigma}_{IC}^{\text{acted}}$
Newey–West t	t_{NW}	t_{NW}^{acted}
Fraction positive	$\Pr(IC_t > 0)$	$\Pr(IC_t^{\text{acted}} > 0)$
Mean coverage	—	$\overline{\text{Coverage}} = \frac{1}{T} \sum_t \frac{N_{\text{act},t}}{N_t}$
Lag-1 autocorr	ρ_1	ρ_1^{acted}
Half-life (periods)	HL	HL^{acted}

Recommended visuals / diagnostics:

- Time series plot of IC_t and IC_t^{acted} (overlay) with shaded NW CI.
- Histogram / density of $\{IC_t\}$ and $\{IC_t^{\text{acted}}\}$; table of summary moments.
- Coverage time series and counts $N^{\text{Long}}, N^{\text{Short}}, N^{\text{Neutral}}$ per period.
- Bootstrap / permutation p-values for $\overline{IC}$ when N_t^{acted} small or imbalanced.

Rolling Hit Rate

Time-series of directional accuracy (explicit handling of neutrals)

$$\text{HitRate}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \mathbf{1}[s_{i,t}^{\text{Train}} r_{i,t}^{\text{Test}} > 0]$$
$$\text{HitRate}_t^{\text{acted}} = \frac{\sum_{i=1}^{N_t} \mathbf{1}[s_{i,t}^{\text{Train}} \neq 0] \mathbf{1}[s_{i,t}^{\text{Train}} r_{i,t}^{\text{Test}} > 0]}{\sum_{i=1}^{N_t} \mathbf{1}[s_{i,t}^{\text{Train}} \neq 0]}$$

- **Time-series aggregation:**

$$\overline{\text{HitRate}} = \overline{\text{HitRate}^{\text{acted}}} =$$

- **Preferred null for directional accuracy:** conduct inference on $\overline{\text{HitRate}^{\text{acted}}}$ and test

$$H_0 : \mathbb{E} \left[\text{HitRate}_t^{\text{acted}} \right] = 0.5,$$

where 0.5 represents

- Neutral positions are excluded because counting them as correct can mechanically inflate the measured hit rate without reflecting successful directional predictions.

Rolling Hit Rate - Aggregation and Inference cont'd

- The sequence $\{\text{HitRate}_t^{\text{acted}}\}_{t=1}^T$ may be **serially correlated** and/or **heteroscedastic**. Newey-West adjusts the estimator of the sample mean for both.
- **Newey-West variance:** let $x_t = \text{HitRate}_t^{\text{acted}}$, $\bar{x} = \overline{\text{HitRate}^{\text{acted}}}$
The Newey-West estimate of the variance of the sample mean is

$$\widehat{\text{Var}}_{\text{NW}}(\bar{x}) = \frac{1}{T} \left[\hat{\gamma}_0 + 2 \sum_{\ell=1}^L \left(1 - \frac{\ell}{L+1} \right) \hat{\gamma}_{\ell} \right],$$

where $\hat{\gamma}_{\ell} = \frac{1}{T} \sum_{t=\ell+1}^T (x_t - \bar{x})(x_{t-\ell} - \bar{x})$ is the estimated autocovariance at lag ℓ .

- **Newey-West test for mean hit rate:**

$$t_{\text{NW}} = \frac{\overline{\text{HitRate}^{\text{acted}}} - 0.5}{\sqrt{\widehat{\text{Var}}_{\text{NW}}(\overline{\text{HitRate}^{\text{acted}}})}}.$$

- The truncation lag L determines how many lagged autocovariances are incorporated; select via rule-of-thumb/data-driven.

- **Per-period exact test:** for each period t , let $N_t^{\text{acted}} = \sum_{i=1}^{N_t} \mathbf{1}[s_{i,t}^{\text{Train}} \neq 0]$, and compute a Binomial($N_t^{\text{acted}}, 0.5$) p-value to flag unusually strong or weak months.
- **Resampling robustness:** block bootstrap over t (to preserve serial dependence), or permutation tests that shuffle realized returns across assets within each t , provide empirical p-values when serial correlation or small counts are concerns.

Rolling Hit Rate — Persistence Diagnostics

Is directional accuracy persistent or transitory?

Persistence diagnostic	Definition / Interpretation
Fraction of positive periods	$\Pr(\text{HitRate}_t^{\text{acted}} > 0.5)$ Frequency with which directional accuracy exceeds random guessing
Lag-1 autocorrelation	$\rho_1^{\text{HitRate}} = \text{corr}(\text{HitRate}_t^{\text{acted}}, \text{HitRate}_{t+1}^{\text{acted}})$ Short-run persistence of directional accuracy
Half-life (periods)	$\text{HalfLife}_{\text{HitRate}} = -\frac{\ln 2}{\ln \rho_1^{\text{HitRate}} }$ Time required for abnormal accuracy to decay by 50%
Mean coverage	$\overline{\text{Coverage}} = \frac{1}{T} \sum_{t=1}^T \frac{N_t^{\text{acted}}}{N_t}$ Fraction of assets with non-zero positions

Hit Rate — Diagnostics & Visuals

Focus on acted signal; show stability, significance, and participation

Recommended plots / diagnostics

- Time series: overlay $\text{HitRate}_t^{\text{acted}}$ and Coverage_t (dual axis); mark months with Binomial p-value < 0.05 .
- Rolling estimate: 6–12 month rolling mean of $\text{HitRate}_t^{\text{acted}}$ with Newey–West CI shading.
- Heatmap of per-period Binomial p-values (months $\times$ years) to flag exceptional windows.
- Distribution: histogram / density of $\{\text{HitRate}_t^{\text{acted}}\}$ with block-bootstrap CI for $\overline{\text{HitRate}}^{\text{acted}}$.
- Scatter: $\text{HitRate}_t^{\text{acted}}$ vs $\text{IC}_t^{\text{acted}}$; annotate correlation and points with low coverage.
- Stability: plot lag-1 autocorr and implied half-life (annotate numeric value).

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Anatomy of a Strategy

Strategy converts signal into investments

- ① Timing:
- ② Position Sizing:

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Position Timing

Converting raw signal s into a traded position s^{pos}

Raw ML signal:

$s_{i,t} \in \{-1, 0, +1\}$ (what the model suggests at time t)

Example:

$$s_{i,\text{Jan}} = +1, \quad s_{i,\text{Feb}} = 0$$

- Model says:
- Naively:
- In practice:

Solution: introduce a *traded signal*

$$s_{i,t}^{pos} \in \{-1, 0, +1\}$$

- $s_{i,t}^{pos}$ converts raw signal $s_{i,t}$ into an actual position

Interpreting $s_{i,t}^{pos}$

The trading layer on top of the signal

- The signal $s_{i,t}$ is evaluated at each time t
- Once a position is opened, it is typically *held*
- The position is closed only when an exit rule is triggered

Common exit trigger types:

- **Time-based:**
- **Asset-based:**
- **State-based:**

Appendix: Formal View of Position Timing

A state-based representation of how signals become positions

$$(s_t^{\text{pos}}, h_t, \tau_t, e_t) = \mathcal{G}(s_t, h_{t-1}, \tau_{t-1}, X_t, Z_t; \Theta)$$

- $s_t = [s_{1,t}, \dots, s_{N,t}]^\top$: raw signals produced by the model at time t
- $s_t^{\text{pos}} \in \{-1, 0, +1\}^N$: positions held over $[t, t+1)$
- $h_{t-1} \in \{0, 1\}^N$: indicators for whether a position is already open
- $\tau_{t-1} \in \mathbb{N}^N$: holding durations entering time t
- e_t : exit indicator / reason (only nonzero when an exit occurs)
- Θ : timing and exit parameters (holding limits, stops, filters)

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Raw Weights: general mapping

Map timing-adjusted signals into signed pre-normalized weights

Let $s_t^{pos} = [s_{1,t}^{pos}, \dots, s_{N,t}^{pos}]^\top$ denote the timing-adjusted signal. We define the *raw* weight vector at time t as a (possibly elementwise) mapping:

$$w_t^{\text{raw}} = \Phi(s_t^{pos}, X_t, Z_t; \Theta_\Phi)$$

where

- $X_t = \{X_{i,t}\}_{i=1}^N$ are asset-level features (e.g.
- Z_t are contextual / cross-sectional features (e.g.
- Θ_Φ : parameters of the sizing map.

Properties / conventions:

- w_t^{raw} is a signed, pre-normalized weight (an intermediate exposure); it does not by itself satisfy portfolio constraints and must be transformed by scaling/optimization to produce tradable weights.
- Typically $w_{i,t}^{\text{raw}} = 0$ when $s_{i,t}^{pos} = 0$ (no action).
- The mapping Φ should be deterministic for reproducibility in backtests.

Raw Example 1: Equal-weight per acted asset

Simple, non-parametric sizing: equal exposure to every active name

Let the acted set be $\mathcal{H}_t = \{i : s_{i,t}^{pos} \neq 0\}$ and $N_t^{\text{acted}} = |\mathcal{H}_t|$. A minimal raw weighting is

$$w_{i,t}^{\text{raw}} =$$

This expresses direction only. A simple cross-sectionally normalized (still pre-portfolio) version is

$$w_{i,t}^{\text{raw}} = \frac{s_{i,t}^{pos}}{N_t^{\text{acted}}}.$$

Notes:

- Very low model risk and easy interpretation.
- Appropriate when the strategy emphasizes selection over sizing.
- Downstream scaling converts these into investable portfolio weights.

Raw Example 2: Volatility-scaled (risk-aware) raw weights

Scale directional intent by inverse realized volatility

Let $\sigma_{i,t}$ be an estimate of asset i 's volatility (e.g. 20-day realized). A common raw mapping is

$$w_{i,t}^{\text{raw}} =$$

Practical adjustment:

$$w_{i,t}^{\text{raw}} \leftarrow \text{clip}(w_{i,t}^{\text{raw}}, -w_{\text{max}}^{\text{raw}}, w_{\text{max}}^{\text{raw}})$$

to prevent extreme pre-weights from very low $\sigma_{i,t}$.

Notes:

- Equalizes per-name risk contribution *before* portfolio normalization.
- Improves stability of downstream scaling in heterogeneous universes.
- Conceptually distinct from full portfolio-level risk parity.

Raw Example 3: Forecast-magnitude raw weights

Size positions by expected return magnitude

Suppose the model produces a one-step-ahead return forecast $\hat{r}_{i,t+1}$.

Define the raw weight

$$w_{i,t}^{\text{raw}} = s_{i,t}^{\text{pos}} \cdot g(\hat{r}_{i,t+1}), \quad g(\hat{r}) \geq 0.$$

Common choices for $g(\cdot)$:

$$g_1(\hat{r}) = \quad g_2(\hat{r}) = \quad g_3(\hat{r}) =$$

Notes:

- Direction is determined by
- Magnitude reflects the model's expected return signal.
- Most natural when forecasts are well-calibrated and comparable cross-sectionally.

Raw Example 4: Confidence-based raw weights

Size positions by model confidence rather than return magnitude

Suppose the model produces a nonnegative confidence score $\pi_{i,t} \geq 0$ (e.g. class probability, margin, or posterior state probability). Define

$$w_{i,t}^{\text{raw}} = s_{i,t}^{\text{pos}} \cdot \pi_{i,t}^{\alpha}, \quad \alpha \in (0, 1].$$

Notes:

- Direction is determined by
- $\pi_{i,t}$ controls
- More robust than return-based sizing when magnitude forecasts are noisy.

Scaled Weights (w^{scaled})

Make raw weights investable via normalization and mechanical constraints

$$w_t^{\text{scaled}} = \mathcal{N}(w_t^{\text{raw}}; \mathcal{C}_t) = \Pi_{\mathcal{C}_t}(\mathcal{S}(w_t^{\text{raw}}))$$

where

- $\mathcal{S}(\cdot)$ is a normalization operator (e.g. gross or net scaling),
- $\Pi_{\mathcal{C}_t}(\cdot)$ enforces feasibility with respect to a set of mechanical constraints $\mathcal{C}_t$,
- $\mathcal{C}_t$ may include: budget constraints, leverage limits, single-name or sector caps, long/short splits, liquidity rules.

Common normalization choices:

$$\mathcal{S}_{L1}(w) = \frac{w}{\sum_i |w_i|}$$

Long-short (gross exposure)

$$\mathcal{S}_{\text{net}}(w) = \frac{w}{\sum_i w_i}$$

Long-only (net exposure)

Normalization and constraint rules should be consistent across strategies so the portfolio layer can combine them meaningfully. Scaling of w^{raw} may set *relative* exposure across assets; w^{scaled} enforces *absolute* budget and feasibility constraints.

Example: L1 normalization with long/short budgets

Define total raw long and short mass:

$$G_t^+ = \sum_{i: w_{i,t}^{\text{raw}} > 0} |w_{i,t}^{\text{raw}}|, \quad G_t^- = \sum_{i: w_{i,t}^{\text{raw}} < 0} |w_{i,t}^{\text{raw}}|$$

Choose long and short budgets $B_L \geq 0$ and $B_S \geq 0$. Define scaled weights:

$$w_{i,t}^{\text{scaled}} = \begin{cases} B_L \frac{|w_{i,t}^{\text{raw}}|}{G_t^+}, & w_{i,t}^{\text{raw}} > 0, \\ -B_S \frac{|w_{i,t}^{\text{raw}}|}{G_t^-}, & w_{i,t}^{\text{raw}} < 0, \\ 0, & w_{i,t}^{\text{raw}} = 0 \end{cases}$$

Properties:

- Gross exposure = $B_L + B_S$; net exposure = $B_L - B_S$
- Preserves relative raw exposures within each side.
- Single-name or sector caps can be enforced by clipping and re-normalizing per side.

Example: Volatility-aware scaling

Normalize raw weights to control ex-ante volatility under diagonal risk

Start from raw weights $w_{i,t}^{\text{raw}}$ and per-asset volatility estimates $\sigma_{i,t}$. Define volatility-adjusted pre-weights:

$$\tilde{w}_{i,t} =$$

Choose a gross exposure target $G > 0$ and scale via L_1 normalization:

$$w_{i,t}^{\text{scaled}} = \frac{\tilde{w}_{i,t}}{\sum_j |\tilde{w}_{j,t}|} \cdot G.$$

Interpretation:

- Preserves directional intent and relative raw strength.
- Equalizes exposure to per-asset volatility (ignoring correlations).
- Controls portfolio gross exposure mechanically.

G sets the target gross exposure: $\sum_i |w_{i,t}^{\text{scaled}}| = G$.
a portfolio-level optimization step and belongs in the w^P layer.

Outline

1 Overview

2 Signals

3 Strategy

- Timing
- Position Sizing
- Strategy Evaluation

4 Portfolio

Strategy Evaluation

Standalone performance of a fully specified trading rule

- Input:
- Output:
- Scope: performance of one strategy at a time; no cross-strategy optimization or global constraints here.

From Scaled Weights to Strategy Returns

$$r_{t+1}^{\text{strat}} = \sum_{i=1}^N w_{i,t}^{\text{scaled}} r_{i,t+1}$$

where

- $w_{i,t}^{\text{scaled}}$: scaled weights or exposures at time
- $r_{i,t+1}$: realized simple asset returns over $[t, t+1)$

Include transaction costs / slippage (example):

$$r_{t+1}^{\text{net}} = \sum_i w_{i,t}^{\text{scaled}} r_{i,t+1} - c \sum_i |w_{i,t} - w_{i,t-1}|$$

where c is per-unit round-trip cost (or a function of turnover).

Notes

- Weights are *ex-ante*; returns are *realized*; align timestamps carefully.
- If your scaling uses gross exposure G or volatility target, it should already be embedded in $w_{i,t}$.

Performance Metrics: Four Categories

The following are a small sample of performance metrics available

1. Absolute Return Measures — “Did it make money?”



2. Risk Measures — “How volatile / tail-risky?”



Performance Metrics (cont.)

Relative, statistical, and practical metrics

3. Risk-Adjusted Measures — “Was return worth the risk?”

- Sharpe ratio:
- Sortino ratio
- Calmar ratio:
- Information Ratio

4. Statistical Reliability Measures — “Skill or noise?”

-
- Newey–West adjusted Sharpe / standard errors
- Bootstrap confidence intervals for CAGR, Sharpe, MaxDrawdown
- Deflated Sharpe (account for multiple trials / backtest overfitting)
- Subsample / rolling stability (e.g., rolling Sharpe, rolling IC)

Outline

- 1 Overview
- 2 Signals
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Portfolios

A portfolio is a combination of strategies subject to global constraints

- Each *strategy* $k = 1, \dots, K$ produces a return stream $r_t^{\text{strat},k}$.
- A *portfolio* combines these strategies using weights $u_t^{\text{strat},k}$
- Portfolio return r_t^p is a weighted combination of $\{r_t^{\text{strat},k}\}$
- Strategy weights u_t^{strat} are chosen to optimize an objective (e.g. risk, return, or risk-adjusted performance) subject to *strategy-level* and *asset-level* constraints
- Although optimization occurs at the strategy level, the portfolio ultimately holds *assets*. Asset weights w_t^p are implied by:
 - each strategy's internal asset weights, and
 - the portfolio's allocation across strategies

Portfolio Components

Strategy construction (asset level):

$$r_t^{\text{strat},k} = \sum_{i=1}^N w_{i,t}^{\text{scaled},k} r_{i,t}, \quad k = 1, \dots, K$$

Portfolio construction (strategy level):

$$r_t^p =$$

Implied portfolio asset weights:

$$w_{i,t}^p =$$

Equivalently, in matrix form:

$$w_t^p = W_t^{\text{scaled}} u_t^{\text{strat}}$$

Portfolio Formation — Optimization

$$\begin{aligned} u_t^* = \arg \max_u \mathbb{E}_t \left[U \left(\sum_{k=1}^K u_t^k r_t^{\text{strat},k} \right) \right] &- \mathbb{E}_t [\ell(\Delta w_t^P(u))] \\ \text{s.t. } A w_t^P \leq b, \quad A_{\text{eq}} w_t^P &= b_{\text{eq}} \\ C u_t \leq d, \quad C_{\text{eq}} u_t &= d_{\text{eq}} \\ w_t^P &= W_t^{\text{scaled}} u_t \end{aligned}$$

- $U(\cdot)$: portfolio objective (e.g. variance, mean–variance, expected utility).
- $\ell(\cdot)$: transaction cost function; $\Delta w_t^P(u) = w_t^P - w_{t-1}^P$.
- A, b : asset-level constraint matrices and bounds ($A \in \mathbb{R}^{R_1 \times N}$).
- C, d : strategy-level constraint matrices and bounds ($C \in \mathbb{R}^{R_2 \times K}$).
- W_t^{scaled} : matrix of strategy asset weights ($N \times K$), mapping strategy allocations to asset weights.

Strategy-level Constraints

$K = 2$ strategies; $R_2 = 2$ inequality constraints and 1 equality constraint

• ?

$$u_t^{\text{strat},k} \geq 0, \quad k = 1, 2$$

$$\underbrace{\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}}_{C \in \mathbb{R}^{R_2 \times K}} \underbrace{\begin{bmatrix} u_t^{\text{strat},1} \\ u_t^{\text{strat},2} \end{bmatrix}}_{u_t} \leq \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_d$$

• ?

$$\sum_{k=1}^2 u_t^{\text{strat},k} = 1$$

$$\underbrace{\begin{bmatrix} 1 & 1 \end{bmatrix}}_{C_{\text{eq}} \in \mathbb{R}^{1 \times K}} u_t = \underbrace{\begin{bmatrix} 1 \end{bmatrix}}_{d_{\text{eq}}}$$

Asset-level Constraints

Individual position limits

- ?

$$w_{i,t}^p \leq 5\%, \quad i = 1, \dots, N$$

$$\underbrace{I_N}_{A_1 \in \mathbb{R}^{R_{1,1} \times N}} \underbrace{w_t^p}_{\in \mathbb{R}^N} \leq \underbrace{0.05 \cdot \mathbf{1}_N}_{b_1 \in \mathbb{R}^{R_{1,1}}}$$

Asset-level Constraints (cont'd)

Sector exposure limits

- ?

Define a sector–asset mapping matrix

$$S \in \{0, 1\}^{M \times N}, \quad S_{m,i} = 1 \text{ if asset } i \text{ belongs to sector } m.$$

Then sector-level constraints can be written as:

$$S w_t^p \leq 0.20 \cdot \mathbf{1}_M$$

$$\underbrace{S}_{A_2 \in \mathbb{R}^{R_{1,2} \times N}} \underbrace{w_t^p}_{\in \mathbb{R}^N} \leq \underbrace{0.20 \cdot \mathbf{1}_M}_{b_2 \in \mathbb{R}^{R_{1,2}}}$$

Summary:

$$A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \in \mathbb{R}^{R_1 \times N}, A \in \mathbb{R}^{M \times N} \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \quad A w_t^p \leq b$$

Mean–Variance Optimization (MVO)

MVO as a building block in the signal–strategy–portfolio framework

- Classic mean–variance optimization (MVO), in the Markowitz tradition, produces portfolio weights that trade off expected return against risk using an expected return vector and a covariance matrix.
- In this course framework, MVO is a *building block*:
 - within a *strategy*, MVO may be used to allocate across assets;
 - at the *portfolio level*, MVO may be used to allocate across strategies.
- Importantly, MVO itself does not specify its inputs. Expected returns and covariances must be supplied by upstream signals, models, or estimators.

Mean–Variance Optimization (MVO) cont'd

Practical considerations and implementation details

- When the investable universe is large, MVO can be numerically unstable due to estimation error in large covariance matrices. A common practical response is to *pre-filter* the universe using a signal.
- For example, assets may be ranked by forecasted Sharpe ratio and a selection rule applied:

$$a_i = \begin{cases} 1, & \text{asset selected for optimization,} \\ 0, & \text{asset excluded.} \end{cases}$$

Optimization is then performed only over assets with $a_i = 1$.

- In practice, strategy construction often proceeds in stages:
 - ① Compute provisional weights (e.g. minimum variance or mean–variance) on the filtered universe.
 - ② Rescale or re-optimize to enforce global constraints such as full investment.
 - ③ Impose additional asset–level constraints (e.g. position limits, sector limits) to obtain final portfolio weights w^P .